Name: **Stanislav Anatolyev / Yaroslav Korobka**
Course: **Econometrics II**

Updated: June 18, 2026

1 Nonparametric estimation

Exercise 1.1. Discuss the behavior of a Nadaraya-Watson mean regression estimate when one of i.i.d. observations, say (x_1, y_1), tends to assume extreme values. Specifically, discuss

(a) the case when x_1 is very big,

(b) the case when y_1 is very big.

Solution.

(a) When x_1 is very big, we will have a big influence of this observation when estimating the regression in the right region of the support. With bounded kernels, we may find that no observations fall into the window. With unbounded kernels, the estimated regression line will go almost through (x_1, y_1).

(b) When y_1 is very big, the estimated regression line will have a hump in the region centered at x_1.

Exercise 1.2. Firms produce some product using technology $f(l,k)$. The functional form of f is unknown, although we know that it exhibits constant returns to scale. For a firm i, we observe labor l_i, capital k_i, and output y_i, and the data-generating process takes the form $y_i = f(l_i, k_i) + \varepsilon_i$, where $\mathbb{E}[\varepsilon_i] = 0$ and ε_i is independent of (l_i, k_i). Using random sample $\{y_i, k_i, l_i\}_{i=1}^n$ suggest a nonparametric estimator of $f(l,k)$ which also exhibits constant returns to scale.

Solution. The CRS technology has the property that

$$f(l,k) = kf\left(\frac{l}{k}, 1\right).$$

Hence, the data-generating process is equivalent to

$$\frac{y_i}{k_i} = f\left(\frac{l_i}{k_i}, 1\right) + \frac{\varepsilon_i}{k_i},$$

from which follows that the Nadaraya-Watson estimator can be written as

$$\hat{f}\left(\frac{l}{k}, 1\right) = \frac{\sum_{i=1}^n K_h\left(\frac{l_i}{k_i} - \frac{l}{k}\right)\frac{y_i}{k_i}}{\sum_{i=1}^n K_h\left(\frac{l_i}{k_i} - \frac{l}{k}\right)} \Rightarrow \hat{f}(l,k) = k \cdot \frac{\sum_{i=1}^n K_h\left(\frac{l_i}{k_i} - \frac{l}{k}\right)\frac{y_i}{k_i}}{\sum_{i=1}^n K_h\left(\frac{l_i}{k_i} - \frac{l}{k}\right)}.$$

Exercise 1.3. Recall the Nadaraya-Watson kernel estimator $\hat{g}(x)$ of the conditional mean $g(x) := \mathbb{E}[y|x]$ constructed for a random sample. Show that if $g(x) = c$, where c is some constant, then $\hat{g}(x)$ is unbiased.

Solution. Assume having a random sample $x_1, \dots, x_n$, and collect it as $\mathcal{X}$. Recall that

$$\hat{g}(x) = \frac{\sum_{i=1}^n y_i K_h(x_i - x)}{\sum_{i=1}^n K_h(x_i - x)}.$$

Then,

$$\begin{aligned}
\mathbb{E}[\hat{g}(x)] &= \mathbb{E}[\mathbb{E}[\hat{g}(x)|\mathcal{X}]] \\
&= \mathbb{E}\left[\mathbb{E}\left[\frac{\sum_{i=1}^n y_i K_h(x_i - x)}{\sum_{i=1}^n K_h(x_i - x)}|\mathcal{X}\right]\right] \\
&= \mathbb{E}\left[\frac{\sum_{i=1}^n \mathbb{E}[y_i|\mathcal{X}] K_h(x_i - x)}{\sum_{i=1}^n K_h(x_i - x)}\right] \\
&= c\mathbb{E}\left[\frac{\sum_{i=1}^n K_h(x_i - x)}{\sum_{i=1}^n K_h(x_i - x)}\right] = c.
\end{aligned}$$

First equality uses the law of iterated expectations, the second uses the definition of the $\hat{g}(x)$, the third follows because kernels are functions of $\mathcal{X}$, the forth follows from the definition of the conditional mean.

Exercise 1.4. Suppose we have a random sample $\{x_i\}_{i=1}^n$ and let

$$\hat{F}(x) = \frac{1}{n}\sum_{i=1}^n \mathbb{1}\{x_i \le x\}$$

denote the empirical distribution function of x_i. Consider two density estimators:

- one-sided estimator:
$$\hat{f}_1(x) = \frac{\hat{F}(x+h) - \hat{F}(x)}{h},$$
- two-sided estimator:
$$\hat{f}_2(x) = \frac{\hat{F}(x+h/2) - \hat{F}(x-h/2)}{h}.$$

Show that:

(a) $\hat{F}(x)$ is an unbiased estimator of $F(x)$.

(b) The bias of $f_1(x)$ is $O(h^a)$. Find the value of a.

(c) The bias of $f_2(x)$ is $O(h^b)$. Find the value of b.

Solution. *Please, consult the note on "big-o", "small-o" notation in case derivations are unclear.*

(a) We show the unbiasedness using the fact that $\mathbb{E}[\mathbb{1}\{x_i \le x\}] = F(x)$. Intuition is as follows: recall that $\mathbb{E}[\mathbb{1}\{x_i = x\}] = \mathbb{P}\{x_i = x\}$; intuitively, with inequality instead of equality those probabilities are summing up until the point x, which leads to equivalence with the cumulative distribution function.

Now, the unbiasedness,

$$\mathbb{E}\left[\hat{F}(x)\right] = \mathbb{E}\left[\frac{1}{n}\sum_{i=1}^n \mathbb{1}\{x_i \le x\}\right] = \frac{1}{n}\sum_{i=1}^n \mathbb{E}[\mathbb{1}\{x_i \le x\}] = F(x).$$

(b) We will use the following Taylor approximation of $F(x+h)$ around x:

$$F(x+h) = F(x) + hf(x) + \frac{1}{2}h^2 f'(x) + o(h^2).$$

Substituting it, we have the bias

$$\begin{aligned}
\mathbb{E}\left[\hat{f}_1(x) - f(x)\right] &= \frac{1}{h}\left(F(x+h) - F(x)\right) - f(x) \\
&= \frac{1}{h}\left(F(x) + hf(x) + \frac{1}{2}h^2 f'(x) + o(h^2) - F(x)\right) - f(x) \\
&= \frac{1}{2}hf'(x) + o(h) \Rightarrow a = 1.
\end{aligned}$$

(c) Similarly to the point (b), we expand $F\left(x+\frac{h}{2}\right)$ and $F\left(x-\frac{h}{2}\right)$ around x:

$$F\left(x+\frac{h}{2}\right) = F(x) + \frac{h}{2}f(x) + \frac{1}{2}\left(\frac{h}{2}\right)^2 f'(x) + \frac{1}{6}\left(\frac{h}{2}\right)^3 f''(x) + o(h^3),$$

and

$$F\left(x-\frac{h}{2}\right) = F(x) - \frac{h}{2}f(x) + \frac{1}{2}\left(\frac{h}{2}\right)^2 f'(x) - \frac{1}{6}\left(\frac{h}{2}\right)^3 f''(x) + o(h^3).$$

Then,

$$\begin{aligned}
\mathbb{E}\left[\hat{f}_2(x)\right] - f(x) &= \frac{1}{h}\left(F\left(x+\frac{h}{2}\right) - F\left(x-\frac{h}{2}\right)\right) - f(x) \\
&= \frac{1}{h}\left(F(x) + \ldots + o(h^3) - F(x) + \ldots + o(h^3)\right) \\
&= \frac{1}{h}\left(\frac{2}{6}\left(\frac{h}{2}\right)^3 f''(x)\right) + o(h^2) \\
&= \frac{1}{24}h^2 f''(x) + o(h^2) \Rightarrow b = 2.
\end{aligned}$$

Exercise 1.5. Derive the asymptotic distribution of the Nadaraya-Watson estimator of the density of a scalar random variable x having a continuous distribution, similarly to how the asymptotic distribution of the Nadaraya-Watson estimator of the regression function is derived, under similar conditions. Give interpretation to how your expressions for asymptotic bias and asymptotic variance depend on the shape of the density.

Solution. Recall the Nadaraya-Watson density estimator, $\hat{f}(x) = \frac{1}{n}\sum_{i=1}^{n} K_h(x_i - x)$. During the first lecture you showed that $\mathbb{E}\left[\hat{f}(x)\right] = f(x) + o(1)$ as $h \to 0$. However, this result is not particularly revealing, since we do not see what causes the bias as h approaches (but not yet equal to) zero. To have a better picture, we work on the bias further taking a higher-order expansion of the density around x (the point we are estimating the density function at).

Now,

$$\begin{aligned}
\mathbb{E}\left[\hat{f}(x) - f(x)\right] &= \int \frac{1}{h}K\left(\frac{x_i - x}{h}\right) f(x_i)dx_i - f(x) \\
&\overset{\left\{u=\frac{x_i-x}{h}\right\}}{=} \int K(u) f\left(x + hu\right) du - f(x) \\
&= \int K(u)\left(f(x) + f'(x)hu + \frac{1}{2}f''(x)h^2u^2 + o(h^2)\right) du - f(x) \\
&= f(x)\int K(u)du + f'(x)h\int uK(u)du + \frac{1}{2}f''(x)h^2\int u^2K(u)du + o(h^2) - f(x) \\
&= \frac{1}{2}f''(x)h^2\sigma_K^2 + o(h^2).
\end{aligned}$$

In particular, we use here the fact that $\int K(u)du = 1$ due to normalization of the kernel, and $\int uK(u)du = 0$ due to the kernel symmetry.

What we see here is revealing: the bias is proportional to the second derivative of the density function. The second derivative, which describes the curvature of the density, is negative ($f''(x) < 0$) around the mode of the distribution, and is positive ($f''(x) > 0$) in the tails. That means that the density estimator will be downward-biased around the mode, and upward-biased around the tails. Also note that our results do not contradict the results you have obtained during the lecture: indeed, by putting $h \to 0$, we are getting rid of the bias.

We start computing the variance with

$$\begin{aligned}
\text{var}\left[\hat{f}(x)\right] &= \text{var}\left[\frac{1}{n}\sum_{i=1}^{n} K_h(x_i - x)\right] = \text{var}\left[\frac{1}{n}\sum_{i=1}^{n}\frac{1}{h}K\left(\frac{x_i - x}{h}\right)\right] \\
&= \frac{1}{n^2h^2}\text{var}\left[\sum_{i=1}^{n} K\left(\frac{x_i - x}{h}\right)\right] = \frac{n}{n^2h^2}\text{var}\left[K\left(\frac{x_i - x}{h}\right)\right] = \frac{1}{nh^2}\text{var}\left[K\left(\frac{x_i - x}{h}\right)\right]
\end{aligned}$$

where the forth equality is implied by the i.i.d. assumption. Next I will normalize by nh to get

$$\begin{aligned}
nh\text{var}\left[\hat{f}(x)\right] &= \frac{1}{h}\text{var}\left[K\left(\frac{x_i-x}{h}\right)\right] \\
&= \frac{1}{h}\mathbb{E}\left[K\left(\frac{x_i-x}{h}\right)^2\right] - \frac{1}{h}\left(\mathbb{E}\left[K\left(\frac{x_i-x}{h}\right)\right]\right)^2 \\
&= \frac{1}{h}\mathbb{E}\left[K\left(\frac{x_i-x}{h}\right)^2\right] - \frac{1}{h}\left(h\cdot\frac{1}{h}\mathbb{E}\left[K\left(\frac{x_i-x}{h}\right)\right]\right)^2 \\
&= \frac{1}{h}\mathbb{E}\left[K\left(\frac{x_i-x}{h}\right)^2\right] - h\left(\mathbb{E}\left[\frac{1}{h}K\left(\frac{x_i-x}{h}\right)\right]\right)^2.
\end{aligned}$$

Note that we have already derived the second term (that is squared here). Specifically,

$$\mathbb{E}\left[\hat{f}(x)\right] = \mathbb{E}\left[\frac{1}{h}K\left(\frac{x_i-x}{h}\right)\right] = f(x) + \frac{1}{2}h^2 f''(x) + o(h^2). \tag{1}$$

Now, let's work with the first term. We have

$$\begin{aligned}
\frac{1}{h}\mathbb{E}\left[K\left(\frac{x_i-x}{h}\right)^2\right] &= \frac{1}{h}\int K\left(\frac{x_i-x}{h}\right)^2 f(x_i)dx_i \\
&\overset{\left\{u=\frac{x_i-x}{h}\right\}}{=} \int K(u)^2 f(x+hu)du \\
&= \int K(u)^2\left(f(x)+o(1)\right)du \\
&= f(x)\int K(u)^2 du + o(1) \\
&= f(x)R_K + o(1).
\end{aligned}$$

Putting everything together, we have

$$\begin{aligned}
nh\text{var}\left[\hat{f}(x)\right] &= \frac{1}{h}\mathbb{E}\left[K\left(\frac{x_i-x}{h}\right)^2\right] - h\left(\mathbb{E}\left[\frac{1}{h}K\left(\frac{x_i-x}{h}\right)\right]\right)^2 \\
&= f(x)R_K + o(1) - h\left(f(x)+\frac{1}{2}h^2 f''(x)+o(h^2)\right)^2 \\
&= f(x)R_K + o(1) - \left(h^2 f(x)+\frac{1}{2}h^4 f''(x)+o(h^4)\right)^2 \\
&= f(x)R_K + o(1) - \left(\mathcal{O}(h^2)\right)^2 \\
&= f(x)R_K + o(1) - \mathcal{O}(h^4) \\
&= f(x)R_K + o(1)
\end{aligned}$$

where we denote $\mathcal{O}(h^2) = h^2 f(x) + \frac{1}{2}h^4 f''(x) + o(h^4)$, because indeed by the definition of "big-o",

$$\mathcal{O}(h^2) = \frac{h^2 f(x) + \frac{1}{2}h^4 f''(x) + o(h^4)}{h^2} = f(x) + \frac{1}{2}h^2 f''(x) + o(h^2) \overset{h\to 0}{\to} f(x) < \infty.$$

Dividing $nh\text{var}\left[\hat{f}(x)\right]$ by nh, we obtain the variance of the Nadaraya-Watson density estimator,

$$\text{var}\left[\hat{f}(x)\right] = \frac{f(x)R_K}{nh} + o\left(\frac{1}{nh}\right) = \mathcal{O}\left(\frac{1}{nh}\right).$$

The CLT of the estimator is implied by the Linderberg CLT for heterogeneous random variables (for which we should verify the Linderberg condition; for simplicity we do not, but it does hold). Hence,

$$\sqrt{nh}\left(\hat{f}(x) - f(x)\right) \overset{d}{\to} \mathcal{N}\left(\frac{1}{2}\lambda f''(x), f(x)R_K\right)$$

where $\lambda := \lim_{n\to\infty} \sqrt{nh^5} < \infty \Leftrightarrow h = \mathcal{O}(n^{-1/5})$. *Notice that the convergence rate should correspond inverse-proportionally to the variance.* Remember, that $h \to 0$ in the limit so we should "delete" h from the variance to stabilize the asymptotic distribution. For example, if the convergence rate is $\sqrt{nh^2}$, we have $hf(x)R_K \to 0$ as $h \to 0$. If, on the other hand, the convergence rate is $\sqrt{nh^{-1}}$, $\frac{f(x)R_K}{h^3} \to \infty$ as $h \to 0$. The convergence rate $\sqrt{nh}$ perfectly balances those two extreme cases giving us a possibility of analyzing the non-degenerate distribution.

Exercise 1.6. Analyze carefully the asymptotic properties of the Nadaraya-Watson estimator of a regression function with pefect fit, i.e., when the variance of the error is zero.

Solution. Given the setup, we can infer that uncertainty will be coming from the distribution of x_i only, since $y_i = g(x_i)$. Let's write the Nadaraya-Watson regression estimator, plug in the data-generating process for y_i, and rewrite:

$$\hat{g}(x) = \frac{\sum_{i=1}^{n} K\left(\frac{x_i - x}{h}\right) y_i}{\sum_{i=1}^{n} K\left(\frac{x_i - x}{h}\right)} = \frac{\sum_{i=1}^{n} K\left(\frac{x_i - x}{h}\right) g(x_i)}{\sum_{i=1}^{n} K\left(\frac{x_i - x}{h}\right)}$$

$$\Rightarrow \hat{g}(x) - g(x) = \frac{\frac{1}{nh}\sum_{i=1}^{n} K\left(\frac{x_i - x}{h}\right) (g(x_i) - g(x))}{\frac{1}{nh}\sum_{i=1}^{n} K\left(\frac{x_i - x}{h}\right)}.$$

We already know that the denominator is the Nadaraya-Watson kernel density estimator. Hence, by previous results,

$$\hat{f}(x) := \frac{1}{nh}\sum_{i=1}^{n} K\left(\frac{x_i - x}{h}\right) \xrightarrow{p} f(x) \neq 0.$$

Given that the probability limit of the denominator is not zero, it is sufficient to show that the numerator is not zero either to have a tractable limiting distribution.

We now focus on the nominator, which is, however, equal to $\hat{q}_2(x)$ term that we have worked with before (during lectures). Specifically, with

$$\hat{q}_2(x) := \frac{1}{nh}\sum_{i=1}^{n} K\left(\frac{x_i - x}{h}\right) (g(x_i) - g(x)) \tag{2}$$

you have showed that

$$\mathbb{E}\left[\hat{q}_2(x)\right] = h^2 B(x) f(x) \sigma_K^2 + o(h^2), \quad \text{var}\ \left[\hat{q}_2(x)\right] = o\left(\frac{1}{nh}\right),$$

with $B(x) := \frac{g''(x)}{2} + \frac{g'(x)f'(x)}{f(x)}$. Equation (2) is the average of random variables

$$\zeta_i(x) := \frac{1}{h} K\left(\frac{x_i - x}{h}\right) (g(x_i) - g(x)).$$

We now focus on the expectation and the variance of $\zeta_i(x)$ to derive its distribution, and once we have it, we could derive the distribution of $\hat{g}(x)$ as well.

Now,

$$
\begin{aligned}
\mathbb{E}\left[\zeta_i(x)^2\right] &= \frac{1}{h^2}\int K\left(\frac{x_i-x}{h}\right)^2 (g(x_i)-g(x))^2 f(x_i)dx_i \\
&\overset{\left\{u=\frac{x_i-x}{h}\right\}}{=} \frac{1}{h}\int K(u)^2 \left(g(x+hu)-g(x)\right)^2 f(x+hu)du \\
&= \frac{1}{h}\int K(u)^2 \left(g'(x)hu+o(h)\right)^2 \left(f(x)+o(1)\right) du \\
&= \frac{1}{h}\int \left(g'(x)hu+o(h)\right)^2 K(u)^2 f(x)du + o(h) \\
&= hg'(x)^2 f(x)\int u^2K(u)^2du + o(h) \\
&= hg'(x)^2 f(x)\Psi_K^2 + o(h)
\end{aligned}
$$

with $\Psi_K^2 := \int u^2K(u)^2du$. Similarly,

$$
\begin{aligned}
\mathbb{E}\left[\zeta_i(x)\right] &= \int \frac{1}{h}K\left(\frac{x_i-x}{h}\right) (g(x_i)-g(x)) f(x_i)dx_i \\
&\overset{\left\{u=\frac{x_i-x}{h}\right\}}{=} \int K(u) \left(g(x+hu)-g(x)\right) f(x+hu)du \\
&= \int K(u)\cdot o(1)(f(x)+f'(x)hu)du = o(h).
\end{aligned}
$$

Together, the variance is $\operatorname{var}\left[\zeta_i(x)\right] = \mathbb{E}\left[\zeta_i(x)^2\right] - \mathbb{E}[\zeta_i(x)]^2$ or

$$hg'(x)^2f(x)\Psi_K^2 + o(h) + o(h^2) = hg'(x)^2f(x)\Psi_K^2 + o(h).$$

By some form of the CLT (notice the change of the convergence rate), we obtain the limiting distribution of ζ_i,

$$\sqrt{nh^{-1}}\left(\frac{1}{nh}\sum_{i=1}^{n}K\left(\frac{x_i-x}{h}\right)(g(x_i)-g(x)) - h^2B(x)\sigma_K^2f(x)+o(h^2)\right) \xrightarrow{d} \mathcal{N}\left(0, g'(x)^2f(x)\Psi_K^2\right).$$

We are finally ready to derive the limiting distribution of $\hat{g}(x)$. So by the CLT and Slustky's theorem,

$$\sqrt{nh^{-1}}(\hat{g}(x)-g(x)) \xrightarrow{d} \mathcal{N}\left(\phi\sigma_K^2B(x), \frac{g'(x)^2}{f(x)}\Psi_K^2\right)$$

with $\phi := \lim_{n\to\infty}\sqrt{nh^3}$.

Exercise 1.7. This problem illustrates the use of a difference operator in nonparametric estimation with i.i.d. data. Suppose there is a scalar variable z that takes values on a bounded support. For simplicity, let z be deterministic and compose a uniform grid on the unit interval $[0,1]$. The other variables are i.i.d. Assume that for the function $g(\cdot)$ below the following Lipschitz condition is satisfied:

$$|g(u)-g(v)| \leq G\,|u-v|$$

for some constant G.

1. Consider a nonparametric regression of y on z:

$$y_i = g(z_i) + e_i, \quad i = 1,\dots,n,$$

where $\mathbb{E}[e_i|z_i] = 0$. Let the data $\{(z_i,y_i)\}_{i=1}^n$ be ordered so that the zs are in increasing order. A first difference transformation results in the following set of equations:

$$y_i - y_{i-1} = g(z_i) - g(z_{i-1}) + e_i - e_{i-1}, \quad i = 2,\dots,n. \tag{3}$$

The target is to estimate $\sigma^2 := \mathbb{E}\left[e_i^2\right]$. Propose its consistent estimator based on the first-difference transformed regression (3). Prove consistency of your estimator.

2. Consider the following partially linear regression of y on x and z:

$$y_i = x_i'\beta + g(z_i) + e_i, \quad i = 1,\dots,n, \tag{4}$$

where $\mathbb{E}\left[e_i|x_i,z_i\right] = 0$. Let the data $\{x_i,z_i,y_i\}_{i=1}^n$ be ordered so that zs are in increasing order. The target is to nonparametrically estimate $g(z)$. Propose its consistent estimator using the first-difference transformation of (4).

Solution.

1. Start with the squared average,

$$\begin{aligned}\frac{1}{n-1}\sum_{i=2}^{n}(y_i-y_{i-1})^2 &= \frac{1}{n-1}\sum_{i=2}^{n}(g(z_i)-g(z_{i-1}))^2+\frac{1}{n-1}\sum_{i=2}^{n}(e_i-e_{i-1})^2 \\ &\quad -\frac{2}{n-1}\sum_{i=2}^{n}(g(z_i)-g(z_{i-1}))(e_i-e_{i-1}).\end{aligned} \tag{5}$$

The logic behind the estimator (5) is that we hope terms with $g(z_i)-g(z_{i-1})$ will disappear due to the smoothness of the function (that asymptotically the difference will converge to zero).

Since z_i compose a uniform grid and are in increasing order, we know that $z_i - z_{i-1} = \dfrac{1}{n-1}$, and we can find the limit of the first term using the Lipschitz condition as

$$\begin{aligned}\left|\frac{1}{n-1}\sum_{i=2}^{n}(g(z_i)-g(z_{i-1}))^2\right| &\le \frac{G^2}{n-1}\sum_{i=2}^{n}(z_i-z_{i-1})^2 \\ &= \frac{G^2}{n-1}\frac{n-1}{(n-1)^2}=\frac{G^2}{(n-1)^2}\to 0\end{aligned}$$

as $n\to\infty$. We can similarly use the Lipschitz condition for the third term,

$$\begin{aligned}\left|\frac{2}{n-1}\sum_{i=2}^{n}(g(z_i)-g(z_{i-1}))(e_i-e_{i-1})\right| &\le \frac{2G}{n-1}\sum_{i=2}^{n}|z_i-z_{i-1}|\,|e_i-e_{i-1}| \\ &= \frac{2G}{(n-1)^2}\sum_{i=2}^{n}|e_i-e_{i-1}| \\ &\le \frac{2G}{(n-1)^2}\sum_{i=2}^{n}(|e_i|+|e_{i+1}|) \\ &= \frac{2G}{n-1}\frac{1}{n-1}\sum_{i=2}^{n}(|e_i|+|e_{i-1}|)\xrightarrow{p}\frac{2G}{n-1}\cdot 2\mathbb{E}\left[|e_i|\right]\to 0\end{aligned}$$

as $n\to\infty$. The first inequality uses the Lipschitz condition of the function, the second inequality uses the triangular inequality, and the probability limit is followed by the law of large numbers and i.i.d. data. *Note that* $\mathbb{E}[|e_i|]\neq 0$*!* We assume that $\mathbb{E}[e_i|z_i]=0$, but the absolute value takes into account only positive values of the random variable.

The second term results in the variance we are after:

$$\frac{1}{n-1}\sum_{i=2}^{n}(e_i-e_{i-1})^2=\frac{1}{n-1}\sum_{i=2}^{n}e_i^2-2e_ie_{i-1}+e_{i-1}^2\xrightarrow{p}2\mathbb{E}[e_i^2]=2\sigma.$$

Hence, the consistent estimator of the variance is

$$\hat{\sigma}^2=\frac{1}{2}\frac{1}{n-1}\sum_{i=2}^{n}(y_i-y_{i-1})^2.$$

2. From the first-difference regression we have

$$\Delta y_i=\Delta x_i'\beta+\Delta g(z_i)+\Delta e_i$$

where Δ denotes the difference parameter. Consider an estimator of β,

$$\hat{\beta}=\left(\sum_{i=2}^{n}\Delta x_i\Delta x_i'\right)^{-1}\sum_{i=2}^{n}\Delta x_i\Delta y_i. \tag{6}$$

Logic behind (6) is very similar to the one in the previous point: we take the difference and hope that the difference between $g(\cdot)$ function at neighboring points will converge to zero.

To prove consistency formally, we have

$$\hat{\beta} - \beta = \left(\frac{1}{n-1} \sum_{i=2}^{n} \Delta x_i \Delta x_i' \right)^{-1} \left(\frac{1}{n-1} \sum_{i=2}^{n} \Delta x_i \left(\Delta g(z_i) + \Delta e_i \right) \right).$$

The denominator is equal in the limit to some non-zero constant C by the law of large numbers,

$$\frac{1}{n-1} \sum_{i=2}^{n} \Delta x_i \Delta x_i' \xrightarrow{p} C.$$

The product

$$\frac{1}{n-1} \sum_{i=2}^{n} \Delta x_i \Delta e_i \xrightarrow{p} 0$$

due to the law of iterated expectations and $\mathbb{E}[e_i | x_i, z_i] = 0$. The remaining term,

$$\begin{aligned} \left| \frac{1}{n-1} \sum_{i=2}^{n} \Delta x_i \Delta g(z_i) \right| &= \left| \frac{1}{n-1} \sum_{i=2}^{n} \Delta x_i \left(g(z_i) - g(z_{i-1}) \right) \right| \\ &\leq \frac{G}{n-1} \sum_{i=2}^{n} |\Delta x_i| |z_i - z_{i-1}| \\ &= \frac{G}{n-1} \frac{1}{n-1} \sum_{i=2}^{n} |\Delta x_i| \xrightarrow{p} 0 \end{aligned}$$

as $n \to 0$, where we again use the Lipschitz condition. As a result, $\hat{\beta} - \beta \xrightarrow{p} 0$.

Having found a consistent estimator of β, we can apply standard nonparametrics to the following regression:

$$y_i - x_i' \hat{\beta} = g(z_i) + e_i^\star, \quad e_i^\star := e_i + x_i'(\beta - \hat{\beta}).$$

Note that if $\hat{\beta}$ was not consistent, we would be working with a misspecified model since $\mathbb{E}[e_i^\star | x_i, z_i] \neq 0$.

Consider the following estimator (we select the rectangular kernel for simplicity),

$$\begin{aligned} \hat{g}(z) &= \frac{\sum_{i=1}^{n} \left(y_i - x_i' \hat{\beta} \right) \mathbb{1}\{|z_i - z| \leq h\}}{\sum_{i=1}^{n} \mathbb{1}\left\{|z_i - z| \leq h\right\}} \\ &= \frac{\sum_{i=1}^{n} \left(g(z_i) + e_i + x_i'(\beta - \hat{\beta}) \mathbb{1}\{|z_i - z| \leq h\} \right)}{\sum_{i=1}^{n} \mathbb{1}\left\{|z_i - z| \leq h\right\}} \\ &= \frac{\sum_{i=1}^{n} g(z_i) \mathbb{1}\{|z_i - z| \leq h\}}{\sum_{i=1}^{n} \mathbb{1}\left\{|z_i - z| \leq h\right\}} + \frac{\sum_{i=1}^{n} e_i \mathbb{1}\{|z_i - z| \leq h\}}{\sum_{i=1}^{n} \mathbb{1}\{|z_i - z| \leq h\}} + \frac{\sum_{i=1}^{n} x_i'(\beta - \hat{\beta}) \mathbb{1}\{|z_i - z| \leq h\}}{\sum_{i=1}^{n} \mathbb{1}\{|z_i - z| \leq h\}} \end{aligned}$$

We work with the first term now. Notice that

$$\left| \frac{\sum_{i=1}^{n} \left(g(z_i) - g(z) \right) \mathbb{1}\{|z_i - z| \leq h\}}{\sum_{i=1}^{n} \mathbb{1}\left\{|z_i - z| \leq h\right\}} \right| \leq G \left| \frac{\sum_{i=1}^{n} |z_i - z| \, \mathbb{1}\{|z_i - z| \leq h\}}{\sum_{i=1}^{n} \mathbb{1}\left\{|z_i - z| \leq h\right\}} \right| \leq Gh \tag{7}$$

which holds because all differences $|z_i - z|$ are less than h due to the indicator function, so we bound the whole expression by h. From it follows that

$$\frac{\sum_{i=1}^{n} g(z_i) \mathbb{1}\{|z_i - z| \leq h\}}{\sum_{i=1}^{n} \mathbb{1}\left\{|z_i - z| \leq h\right\}} = \frac{\sum_{i=1}^{n} \left(g(z) + g(z_i) - g(z) \right) \mathbb{1}\{|z_i - z| \leq h\}}{\sum_{i=1}^{n} \mathbb{1}\left\{|z_i - z| \leq h\right\}} \leq g(z) + Gh \to g(z)$$

as $h \to 0$ where we use (7).

For the third term, it holds by the law of iterated expectations and of large numbers that

$$\frac{\sum_{i=1}^{n} x_i' \mathbb{1}\{|z_i - z| \leq h\}}{\sum_{i=1}^{n} \mathbb{1}\{|z_i - z| \leq h\}} \xrightarrow{p} \frac{\mathbb{E}\left[x_i' \mathbb{1}\{|z_i - z| \leq h\}\right]}{\mathbb{E}\left[\mathbb{1}\{|z_i - z| \leq h\}\right]} = \mathbb{E}[x_i'],$$

and by $\hat{\beta} - \beta \xrightarrow{p} 0$, we have that

$$\frac{\sum_{i=1}^{n} x_i'(\beta - \hat{\beta}) \mathbb{1}\{|z_i - z| \leq h\}}{\sum_{i=1}^{n} \mathbb{1}\{|z_i - z| \leq h\}} \xrightarrow{p} 0.$$

Finally, for the third term, we have by the law of iterated expectations

$$\frac{\sum_{i=1}^{n} e_i \mathbb{1}\{|z_i - z| \leq h\}}{\sum_{i=1}^{n} \mathbb{1}\{|z_i - z| \leq h\}} \xrightarrow{p} 0.$$

Hence, $\hat{g}(z) \xrightarrow{p} g(z)$.

Exercise 1.8. Let $(x_i, y_i)_{i=1}^n$ be a random sample from the population of (x, y), where x has a discrete distribution with the support $a_{(1)}, \ldots, a_{(k)}$, where $a_{(1)} < \ldots < a_{(k)}$. Having written the conditional expectation $\mathbb{E}[y|x = a_{(j)}]$ in the form that allows to apply the analogy principle, propose an analog estimator $\hat{g}_j$ of $g_j = \mathbb{E}[y|x = a_{(j)}]$ and derive its asymptotic distribution.

Solution. Fix $a_{(j)}$ for $j = 1, \ldots, k$. Then

$$g_j := g_i(a_{(j)}) = \mathbb{E}[y_i|x_i = a_{(j)}] = \frac{\mathbb{E}[y_i \mathbb{1}\{x_i = a_{(j)}\}]}{\mathbb{E}[\mathbb{1}\{x_i = a_{(j)}\}]},$$

which holds because

$$\begin{aligned}
\mathbb{E}[\mathbb{1}\{x_i = a_{(j)}\}] &= \mathbb{E}[\mathbb{1}\{x_i = a_{(j)}\}|x_i = a_{(j)}] \cdot \mathbb{P}\{x_i = a_{(j)}\} \\
&\quad + \mathbb{E}[\mathbb{1}\{x_i = a_{(j)}\}|x_i \neq a_{(j)}] \cdot \mathbb{P}\{x_i \neq a_{(j)}\} \\
&= \mathbb{P}\{x_i = a_{(j)}\},
\end{aligned}$$

and

$$\begin{aligned}
\mathbb{E}[y_i \mathbb{1}\{x_i = a_{(j)}\}] &= \mathbb{E}[y_i \mathbb{1}\{x_i = a_{(j)}\}|x_i = a_{(j)}] \cdot \mathbb{P}\{x_i = a_{(j)}\} \\
&\quad + \mathbb{E}[y_i \mathbb{1}\{x_i = a_{(j)}\}|x_i \neq a_{(j)}] \cdot \mathbb{P}\{x_i \neq a_{(j)}\} \\
&= \mathbb{E}[y_i|x_i = a_{(j)}] \cdot \mathbb{P}\{x_i = a_{(j)}\}.
\end{aligned}$$

Hence, the estimator is

$$\hat{g}_i(a_{(j)}) = \frac{\sum_{i=1}^n y_i \mathbb{1}\{x_i = a_{(j)}\}}{\sum_{i=1}^n \mathbb{1}\{x_i = a_{(j)}\}} \xrightarrow{p} \frac{\mathbb{E}[y_i \mathbb{1}\{x_i = a_{(j)}\}]}{\mathbb{E}[\mathbb{1}\{x_i = a_{(j)}\}]} := g_i(a_{(j)}).$$

To derive the asymptotic distribution, normalize

$$\begin{aligned}
\sqrt{n}(\hat{g}_j - g_j) &= \sqrt{n}\left(\frac{\sum_{i=1}^n y_i \mathbb{1}\{x_i = a_{(j)}\}}{\sum_{i=1}^n \mathbb{1}\{x_i = a_{(j)}\}} - g_j\right) \\
&= \sqrt{n}\left(\frac{\sum_{i=1}^n y_i \mathbb{1}\{x_i = a_{(j)}\} - g_j \sum_{i=1}^n \mathbb{1}\{x_i = a_{(j)}\}}{\sum_{i=1}^n \mathbb{1}\{x_i = a_{(j)}\}}\right) \\
&= \sqrt{n}\left(\frac{\sum_{i=1}^n (y_i - \mathbb{E}[y_i|x_i = a_{(j)}]) \mathbb{1}\{x_i = a_{(j)}\}}{\sum_{i=1}^n \mathbb{1}\{x_i = a_{(j)}\}}\right).
\end{aligned}$$

So by the CLT, the numerator is distributed as

$$\frac{1}{\sqrt{n}}\sum_{i=1}^{n}(y_i - \mathbb{E}[y_i|x_i = a_{(j)}])\mathbb{1}\{x_i = a_{(j)}\} \xrightarrow{d} \mathcal{N}(0,\omega),$$

where ω is the asymptotic variance given by

$$\begin{aligned}\omega &:= \text{var}\,[(y_i - \mathbb{E}[y_i|x_i = a_{(j)}])\mathbb{1}\{x_i = a_{(j)}\}] \\ &= \mathbb{E}[(y_i - \mathbb{E}[y_i|x_i = a_{(j)}])^2|x_i = a_{(j)}] \cdot \mathbb{P}\{x_i = a_{(j)}\} \\ &= \text{var}\,[y_i|x_i = a_{(j)}] \cdot \mathbb{P}\{x_i = a_{(j)}\}.\end{aligned}$$

By the Slutsky's theorem and CMT, we have that

$$\sqrt{n}(\hat{g}_j - g_j) \xrightarrow{d} \mathcal{N}\left(0, \frac{\text{var}\,[y_i|x_i = a_{(j)}]}{\mathbb{P}\{x_i = a_{(j)}\}}\right).$$

2 Discrete choice models

Exercise 2.1. Let X be distibuted as $\Lambda(x) := \frac{e^x}{1+e^x}$. Verify that

a) $\frac{\partial \Lambda(x)}{\partial x} = \Lambda(x)(1 - \Lambda(x))$,

b) $\frac{\partial \log \Lambda(x)}{\partial x} = 1 - \Lambda(x)$,

c) $-\frac{\partial^2 \log \Lambda(x)}{\partial x^2} = \Lambda(x)(1 - \Lambda(x))$.

Lemma 2.2. *For some constant a, we have*

$$\int_{-\infty}^{\infty} ae^{-\varepsilon}\exp\left(-ae^{-\varepsilon}\right)d\varepsilon = 1.$$

Exercise 2.3. Derive the logit from the Gumbel distribution of ε_0 and ε_1.

Exercise 2.4. Consider a latent variable modeled by $y_i^\star = x_i'\beta + \varepsilon_i$, $\varepsilon_i \sim \mathcal{N}(0,1)$. Suppose we observe only $y_i = 1$ if $y_i^\star < U_i$, and $y_i = 0$ if $y_i \geq U_i$, where the upper limit U_i is a known constant for each i and may differ over i.

Exercise 2.5. In 2022, CERGE-EI conducted a small research aimed at studying a relationship between drinking alcohol and the pass rate of the Econometrics II course. Table 1 presents the number of students in each category. Dummy variable d_i takes a value of 1 if a student i has been drinking one drink per day during the final week, 0 otherwise.

Table 1: Data on 100 students

$d_i \setminus y_i$	fail	pass
no beer	24	28
beer	32	16

To answer this research question, CERGE-EI committee decided to specify the model as

$$y_i^\star = \alpha + \beta d_i + \varepsilon_i, \quad i = 1, \ldots, 100,$$

and they observe only

$$y_i = \begin{cases} 1 & \text{if } y_i^\star > 0, \\ 0 & \text{otherwise.} \end{cases}$$

Obtain the maximum-likelihood estimators of α and β specifying the model as logit. Should you drink during the final week?

Exercise 2.6. For the conditional logit,

$$p_{ij} = \frac{e^{x'_{ij}\beta}}{\sum_{\ell} e^{x'_{i\ell}\beta}},$$

show that

a) $\dfrac{\partial p_{ij}}{\partial \beta} = p_{ij}(x_{ij} - \bar{x}_i)$, where $\bar{x}_i := \sum_{l=1}^{m} p_{i\ell} x_{i\ell}$, and

b) $\dfrac{\partial p_{ij}}{\partial x_{ik}} = p_{ij}(\delta_{ijk} - p_{ik})\beta$, where $\delta_{ijk} := \mathbb{1}\{j = k\}$.